



NTSB Accident Investigation Report Analysis

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2/25/26 meeting on Comparisons for Zhang's approach and
My approach

Example 1: Pilot Error/ Unstable approach

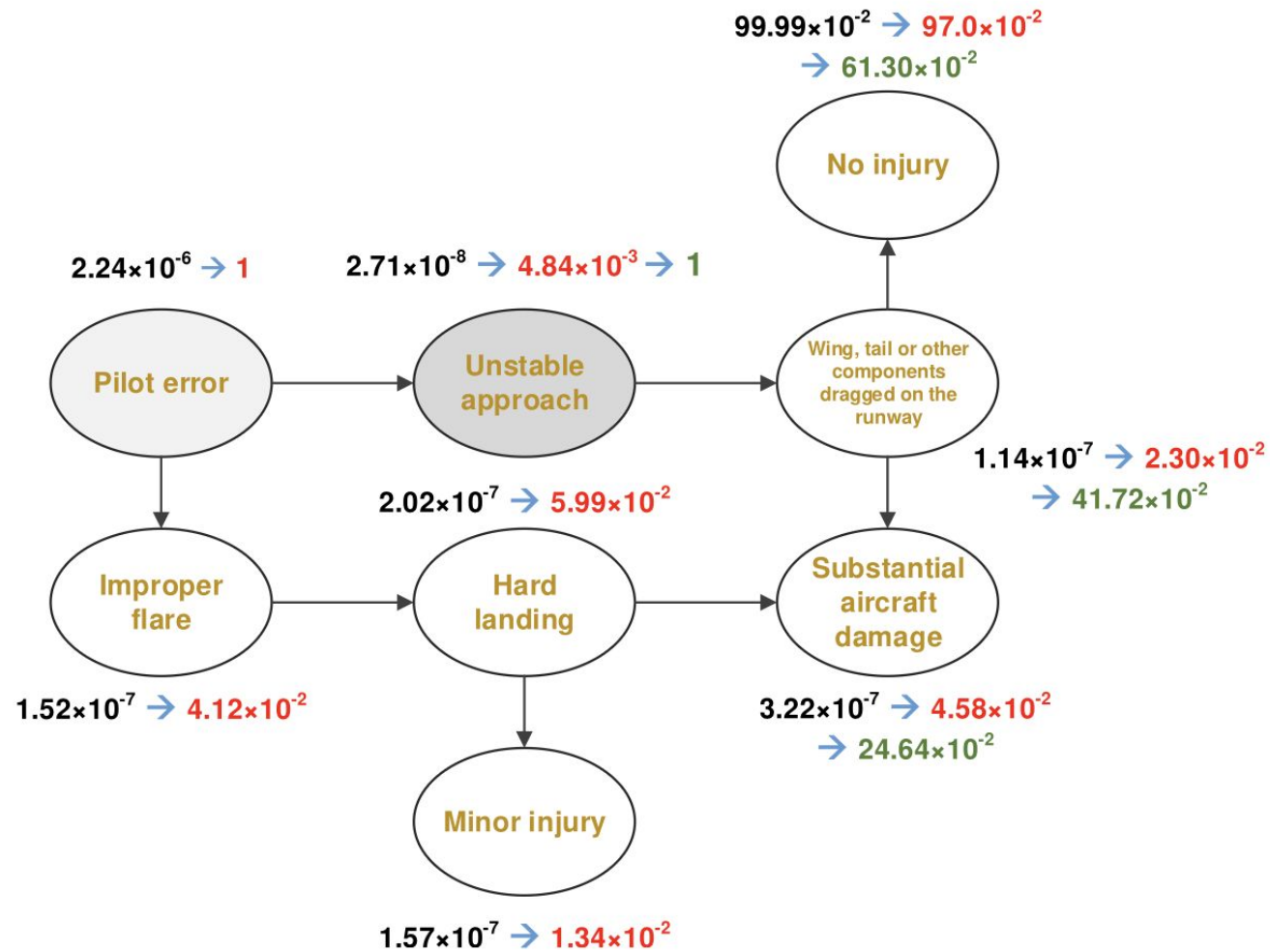


Fig. 12. The influence propagation in the Bayesian network as a result of observing pilot-in-command's erroneous action, where black fonts represent the prior probability of each event, red fonts denote the updated posterior probability after observing the first evidence – pilot-in-command, and green fonts denote the posterior probability after observing the second evidence – unstabilized approach. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Both approaches

His approach:

- Set $P(\text{Pilot Error}) = 1$ manually in GeNie Software

My approach:

Aircraft experienced an unstable approach during landing with improper flare and hard landing resulting in substantial aircraft damage



Incident Clustering Analysis

Clustered 50 incidents into 8 types

Cluster Type	$P(\text{Cluster} \text{Query})$	Incidents	Avg Similarity	Unique Causes
Thrust Reduction	38.5%	19	0.613	27
Investigation Ongoing	21.7%	11	0.598	1
Pilot Error	18.1%	9	0.609	14
Landing Error	11.9%	6	0.600	10
Hydraulic Failure	4.0%	2	0.605	5
Landing Gear Failure	1.9%	1	0.589	2
Bird Strike	1.9%	1	0.584	2
Insufficient Information	0.0%	1	0.584	0



Step 2: Final Root Cause Diagnosis

What's happening here: We combine results from ALL clusters using the chain rule (Law of Total Probability). For each possible cause, we multiply $P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query})$ for every cluster, then sum them up. This gives one final probability that weighs each cluster's contribution by how relevant it is to the query. Click on any cause below to see the full calculation breakdown.

Combined probability using the Total Probability Theorem: $P(C|Q) = \sum P(C|I) \times P(I|Q)$

1. Personnel issues-Action/decision-Action-Incorrect action sequence-Pilot - C

Probability

21.7%

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (1.0000 \times 0.2171) = 0.2171$ (21.7%)

[View cluster-by-cluster breakdown](#)



Investigation Ongoing: $P(C|\text{Cluster})=1.0000 \times P(\text{Cluster}|Q)=0.2171 = 0.2171$

2. Personnel issues-Task performance-Use of equip/info-Aircraft control-Pilot - C

Probability

3.7%

Calculation: $\sum P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0952 \times 0.3848) = 0.0366$ (3.7%)

[View cluster-by-cluster breakdown](#)



Thrust Reduction: $P(C|\text{Cluster})=0.0952 \times P(\text{Cluster}|Q)=0.3848 = 0.0366$



****3. the early reduction of engine thrust that resulted in an excessive sink rate and nose high pitch attitude during the landing flare.**

Probability
2.7%

Contributing to the accident was a significant change in the vertical and horizontal wind components just prior to touchdown.**

Calculation: $\Sigma P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0714 \times 0.3848) = 0.0275 \text{ (2.7\%)}$

View cluster-by-cluster breakdown

Thrust Reduction: $P(C|\text{Cluster})=0.0714 \times P(\text{Cluster}|Q)=0.3848 = 0.0275$

4. Personnel issues-Action/decision-Action-Incorrect action performance-Flight crew - C

Probability
2.7%

Calculation: $\Sigma P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0714 \times 0.3848) = 0.0275 \text{ (2.7\%)}$

View cluster-by-cluster breakdown

Thrust Reduction: $P(C|\text{Cluster})=0.0714 \times P(\text{Cluster}|Q)=0.3848 = 0.0275$

5. Aircraft-Aircraft oper/perf/capability-Performance/control parameters-Pitch control-Incorrect use/operation - C

Probability
1.8%

Calculation: $\Sigma P(\text{Cause}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.0476 \times 0.3848) = 0.0183 \text{ (1.8\%)}$

View cluster-by-cluster breakdown

Thrust Reduction: $P(C|\text{Cluster})=0.0476 \times P(\text{Cluster}|Q)=0.3848 = 0.0183$

Comparison



Ex: Example 1: Pilot Error/ Unstable approach

	Zhang & Maha Paper 2021	Our Approach
Direction	Forward: cause → effect	Backward: symptom → Cause
Input	Set P(Pilot Error) = 1 manually	Query typed by user
Method	Bayesian Network (740 nodes)	Precomputed Embedding + Clustering + chain rule
Tool Required	GeNIe modeler	Web Browser/Streamlit
key output	P(unstable approach) increases from near zero to 4.84×10^{-3}	Thrust Reduction cluster: 38.5%
Top results	Substantial damage: 24.64%	Pilot error as top cause: 21.7%
Handles new queries	No, static network	yes, any natural language input from the same NTSB data

Takeaways:



- **Both agree:** pilot error/crew performance is the central factor
- Zhang starts from known cause, our pipeline starts from observed symptom
- **Limitation:** small clusters can inflate probabilities due to limited data (shared challenge noted by Zhang)

Example 2: (loss of Engine power)

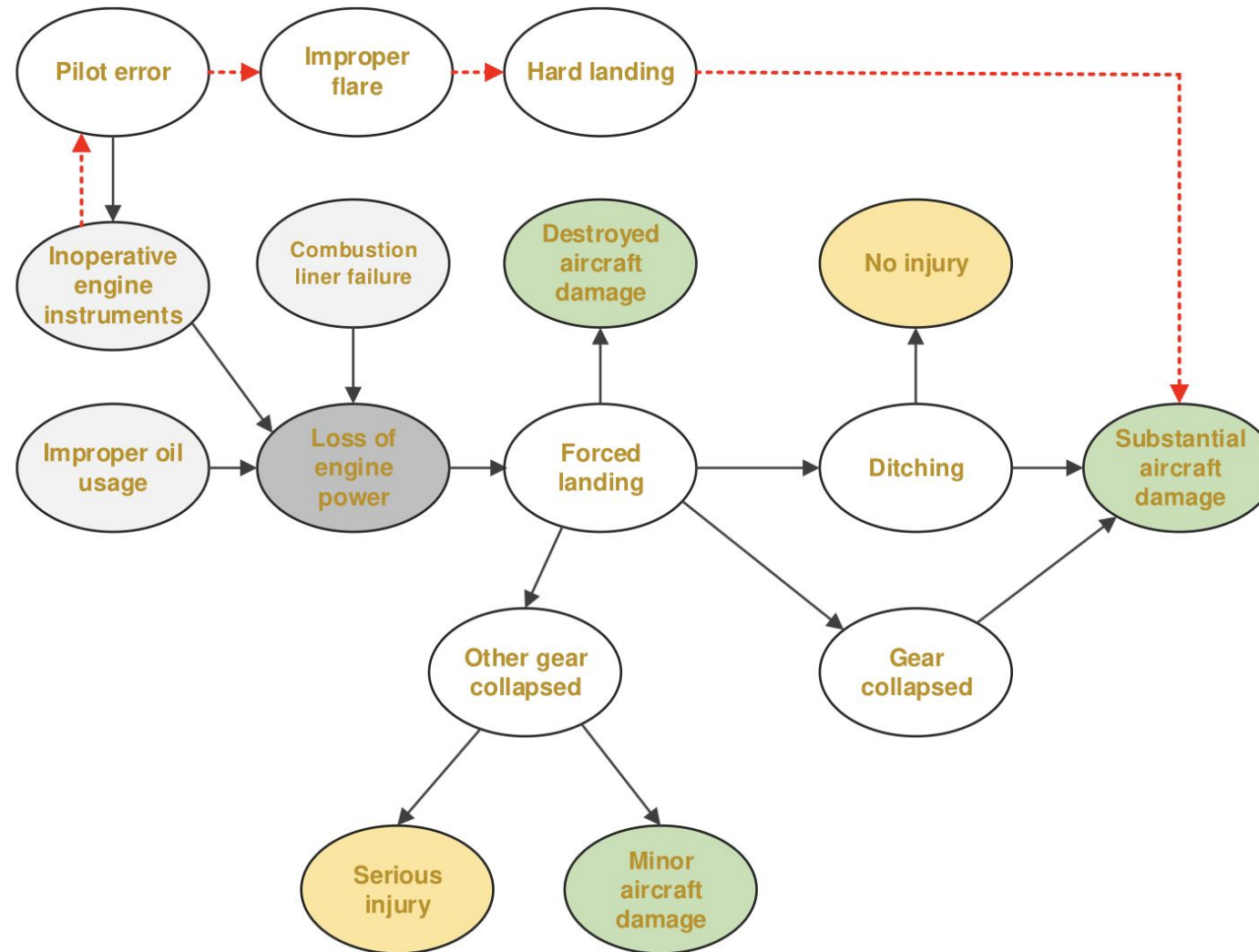


Fig. 13. Influence propagation as a result of the loss of engine power.



Table 9
The posterior probabilities related to each event when observing different evidences.

Evidence	Inoperative engine instruments	Combustion liner failure	Improper oil usage	Inoperative engine instruments & Improper oil usage	Loss of engine power
Loss of engine power	0.95	0.50	0.95	0.99	1
Forced landing	13.57×10^{-2}	7.14×10^{-2}	13.57×10^{-2}	14.71×10^{-2}	14.29×10^{-2}
Ditching	4.37×10^{-3}	2.30×10^{-3}	4.37×10^{-3}	4.57×10^{-3}	4.61×10^{-3}
Gear collapsed	9.60×10^{-3}	2.30×10^{-3}	4.37×10^{-3}	9.82×10^{-3}	5.18×10^{-3}
Other gear collapsed	4.80×10^{-3}	2.30×10^{-3}	4.37×10^{-3}	5.00×10^{-3}	4.66×10^{-3}
Destroyed aircraft	1.33×10^{-2}	2.30×10^{-3}	4.37×10^{-3}	1.35×10^{-2}	5.59×10^{-3}
Substantial aircraft damage	4.60×10^{-2}	3.63×10^{-3}	6.09×10^{-3}	4.63×10^{-2}	1.66×10^{-2}
Minor aircraft damage	9.34×10^{-3}	1.54×10^{-3}	2.92×10^{-3}	9.47×10^{-3}	3.78×10^{-3}
Serious injury	6.23×10^{-2}	7.68×10^{-4}	1.46×10^{-3}	6.23×10^{-2}	8.22×10^{-3}
No injury	94.31×10^{-2}	99.78×10^{-3}	99.58×10^{-3}	94.29×10^{-2}	98.99×10^{-2}

Both approaches



Zhang approach:

- set node “inoperative engine instrument” = 1 in GeNie SOfware

My input:

Prognosis Method

Choose prognosis approach: ?

- ☒ Event Sequence Prediction (Professor's Method)
- ☐ Time-to-Failure (Legacy)

Enter Incident Description / Current Symptom:

Loss of engine power during cruise flight due to inoperative engine instruments

Clustering table



Clustered 45 incidents into 9 types

Incident Type	P(Cluster Query)	Incidents	Sequences
Turbine Blade Failure	43.2%	19	18
Insufficient Information	18.1%	8	0
Ice Ingestion	13.8%	6	1
Engine Fire	12.6%	6	4
Apu Failure	4.1%	2	0
Bird Strike	2.2%	1	1
Electrical Failure	2.0%	1	1
Manufacturing Error	2.0%	1	0
Loss Of Control	2.0%	1	0



Step 3: final event prognosis top results

Step 3: Final Event Prognosis

What's happening here: We combine results from ALL clusters using the chain rule. For each possible next event, we multiply $P(\text{NextEvent}|\text{Cluster}) \times P(\text{Cluster}|\text{Query})$ for every cluster, then sum them up. This gives us one final probability that weighs each cluster's contribution by how relevant it is to the query. The 'Calculation' line shows exactly which clusters contributed and how much.

Combined probability using the Total Probability Theorem:

$$P(E | Q) = \sum P(E | I) \times P(I | Q)$$

1. Initial climb - Powerplant sys/comp malf/fail — Probability: 16.2%

Probability: 16.2%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (1.0000 \times 0.1383) + (0.0556 \times 0.4316) = 0.1623$ (16.2%)

Contributing clusters: 2

Cluster-by-cluster breakdown:

Ice Ingestion: $P(\text{Event}|\text{Cluster})=1.0000 \times P(\text{Cluster}|\text{Q})=0.1383 = 0.1383$

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.0556 \times P(\text{Cluster}|\text{Q})=0.4316 = 0.0240$

Supporting incidents: 20170608X90705, 20020612X00866, 20140922X14716

2. Initial climb - Engine shutdown — Probability: 11.1%

Probability: 11.1%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.5000 \times 0.1258) + (0.1111 \times 0.4316) = 0.1108$ (11.1%)

Contributing clusters: 2

Cluster-by-cluster breakdown:

Engine Fire: $P(\text{Event}|\text{Cluster})=0.5000 \times P(\text{Cluster}|\text{Q})=0.1258 = 0.0629$

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.1111 \times P(\text{Cluster}|\text{Q})=0.4316 = 0.0480$

Supporting incidents: 20170608X90705, 20020917X01767, 20140922X14716

3. Enroute-cruise - Fire/smoke (non-impact) — Probability: 9.6%

Probability: 9.6%

Calculation: $\sum P(\text{Event}|\text{Cluster}) \times P(\text{Cluster}|\text{Query}) = (0.2222 \times 0.4316) = 0.0959$ (9.6%)

Contributing clusters: 1

Cluster-by-cluster breakdown:

Turbine Blade Failure: $P(\text{Event}|\text{Cluster})=0.2222 \times P(\text{Cluster}|\text{Q})=0.4316 = 0.0959$

Supporting incidents: 20170608X90705, 20140922X14716, 20120221X03231

Mult step results



Multi-Step Event Sequence (Bayesian Chain)

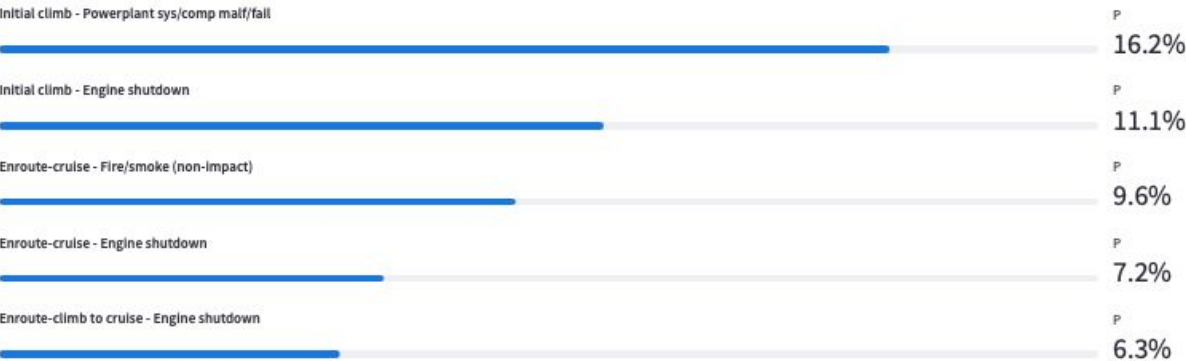
What's happening here: Step 1 predicts the immediate next event. But incidents cascade — A → B → C. To predict Step 2 and Step 3, we use transition probabilities built from consecutive event pairs in the retrieved incidents' sequences. For each Step 1 event, we look at what historically follows it, then weight by how likely that Step 1 event was. This chains the prediction forward through multiple steps.

Extending the prediction forward using transition probabilities:

$$P(Event_t | Q) = \sum P(Event_t | Event_{t-1}) \times P(Event_{t-1} | Q)$$

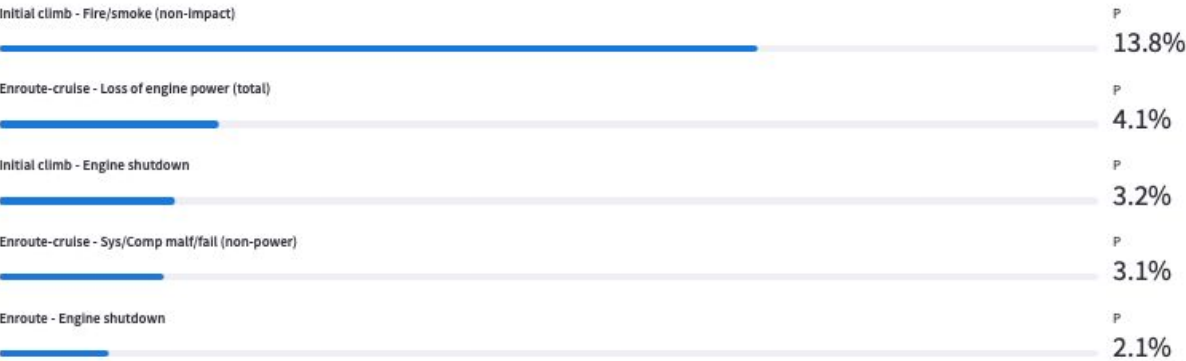
Step 1: Immediate Next Event

These are the same results from Step 3 above — the most likely events to happen immediately after the query symptom.



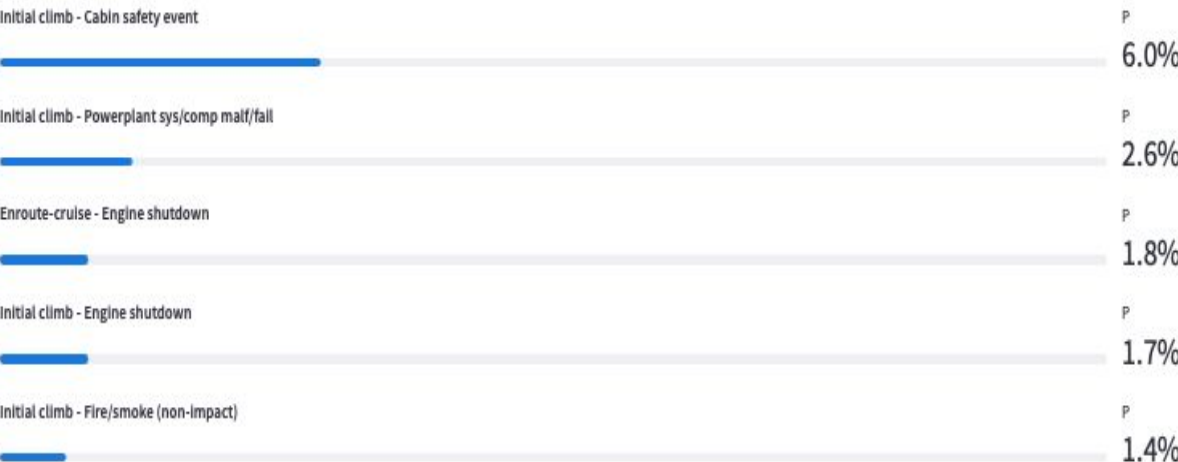
Step 2: What Follows After

These events are predicted by looking at what historically follows Step 1 events in the sequence data, weighted by each Step 1 event's probability.



Step 3: What Follows After

These events are predicted by looking at what historically follows Step 2 events in the sequence data, weighted by each Step 2 event's probability.



Comparisons::



	Zhang & Mahadevan	Our Approach
Direction	Forward: Cause → outcome	Forward: symptom → event sequence
Input	Set node inoperative engine instruments = 1 in Genie	Query typed by user
Method	Bayesian Network forward propagation	Precomputed embeddings + clusters + chain rule + transition probabilities
Tool Required	Genie software	Web browser/ Streamlit
Top Cluster	Single cause node triggered	Turbine Blade failure: 43.2%
Step 1 output	Loss of engine power: 95%	powerplant malfunction: 16.2%, engine shutdown 11.1%,
Step 2 output	forced landing: 13.57%	fire/smoke: 13.8%
Step 3 output	Substantial damage: 4.6%, serious injury: 6.23%	cabin safety event: 6%

takeaways:



- both approaches show the same escalation story
 - engine problem → shutdown → fire → safety/damage outcome
- Zhang's paper predicts final outcomes, while my pipeline predictss the event sequence that leads to those outcomes
- they complement each other
 - Zhang paper tells you how bad it gets, mine tells you the chian of events that gets you there

Current Timeline

- Meeting Jesse Friday
 - Show Streamlit app
 - Show ModernBERT
- First draft comparisons, future tasks, and conclusion
- Talk to Dr. Maha about classes and all today before we finish

